

INDIAN STATISTICAL INSTITUTE, BANGALORE CENTRE
B.MATH - Second Year, 2023-24
Statistics - III, Backpaper Examination, June, 2024
Time: 2 Hours **Total Marks: 50**

1. Let $\mathbf{Y} \sim N_n(\mathbf{0}, \sigma^2 I_n)$. Find the conditional distribution of $\mathbf{Y}'\mathbf{Y}$ given $\mathbf{a}'\mathbf{Y} = 0$ where $\mathbf{a}$ is a non-zero constant vector. [8]

2. Consider the model $\mathbf{Y} = \mathbf{X}\beta + \epsilon$, where $\mathbf{X}_{n \times p}$ has $\mathbf{1}$ as its first column and rank $r \leq p$, and $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$.

(a) If $\hat{\beta}$ is the least squares solution of β , show that $(\hat{\beta} - \beta)' \mathbf{X}' \mathbf{X} (\hat{\beta} - \beta)$ is distributed independently of the residual sum of squares.

(b) Find the maximum likelihood estimator of σ^2 . Is it unbiased?

(c) Explain how the coefficient of determination, R^2 , can be used to check the quality of the fitted linear model. [6 + 6 + 6]

3. Consider the following model:

$$y_1 = \theta + \gamma + \epsilon_1$$

$$y_2 = \theta + \phi + \epsilon_2$$

$$y_3 = 2\theta + \phi + \gamma + \epsilon_3$$

$$y_4 = \phi - \gamma + \epsilon_4,$$

where ϵ_i are uncorrelated having mean 0 and variance σ^2 .

(a) Show that $\gamma - \phi$ is estimable. What is its BLUE?

(b) Find the residual sum of squares. What is its degrees of freedom? [8 + 6]

4. Let Y be a response variable and $X_1, \dots, X_k$ be covariates. Also, let r_i denote the correlation coefficient between Y and X_i , and let R denote the multiple correlation coefficient between Y and $X_1, \dots, X_k$.

(a) Show that $R \geq \max\{|r_i|, 1 \leq i \leq k\}$.

(b) What is the exact relationship between R and r_i 's when $k = 1$? [5+5]